

ORF 309 Homework 3 – Question 4: Love Triangles

Problem Setup

A dating website has n people signed up. For every pair of people, the website independently flips a coin that comes up heads with probability p . If heads, the pair is matched. A **love triangle** is a set of three people who are all mutually matched.

Part (a): Expected Number of Hookups for a Given Individual

Step 1: Identify the random experiment.

Fix a particular person, call them Person 1. There are $n - 1$ other people on the site. For each of these $n - 1$ people, the website independently flips a coin with probability p of heads.

Step 2: Define indicator variables.

For each person $j \in \{2, 3, \dots, n\}$, define:

$$X_j = \begin{cases} 1 & \text{if Person 1 is matched with Person } j \\ 0 & \text{otherwise} \end{cases}$$

Each X_j is a Bernoulli random variable with $P(X_j = 1) = p$, so $E[X_j] = p$.

Step 3: Express total hookups as a sum.

The total number of hookups Person 1 has is:

$$H = \sum_{j=2}^n X_j$$

Step 4: Apply linearity of expectation.

$$E[H] = E\left[\sum_{j=2}^n X_j\right] = \sum_{j=2}^n E[X_j] = \sum_{j=2}^n p = (n-1)p$$

Answer: $E[\text{hookups}] = (n-1)p$

Part (b): Probability of More Than One Hookup

Step 1: Identify the distribution.

From Part (a), $H = \sum_{j=2}^n X_j$ where each X_j is an independent Bernoulli(p). The sum of $n - 1$ independent Bernoulli(p) variables follows a **Binomial distribution**:

$$H \sim \text{Binomial}(n - 1, p)$$

Step 2: Use the complement.

$$P(H > 1) = 1 - P(H = 0) - P(H = 1)$$

Step 3: Compute $P(H = 0)$.

$$P(H = 0) = \binom{n-1}{0} p^0 (1-p)^{n-1} = (1-p)^{n-1}$$

This is the probability that *none* of the $n - 1$ coin flips come up heads.

Step 4: Compute $P(H = 1)$.

$$P(H = 1) = \binom{n-1}{1} p^1 (1-p)^{n-2} = (n-1)p(1-p)^{n-2}$$

This is the probability that exactly one of the $n - 1$ flips comes up heads.

Step 5: Combine.

$$P(H > 1) = 1 - (1-p)^{n-1} - (n-1)p(1-p)^{n-2}$$

Answer: $P(\text{more than one hookup}) = 1 - (1-p)^{n-1} - (n-1)p(1-p)^{n-2}$

Part (c): Expected Number of Love Triangles

Step 1: Enumerate all possible triangles.

A love triangle requires choosing 3 people out of n . The number of ways to do this is:

$$\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$$

Step 2: Define indicator variables for each triple.

For each triple of people $\{i, j, k\}$, define:

$$T_{ijk} = \begin{cases} 1 & \text{if } \{i, j, k\} \text{ form a love triangle} \\ 0 & \text{otherwise} \end{cases}$$

The total number of love triangles on the site is:

$$L = \sum_{\{i,j,k\}} T_{ijk}$$

where the sum is over all $\binom{n}{3}$ triples.

Step 3: Compute $P(T_{ijk} = 1)$ for a given triple.

For $\{i, j, k\}$ to be a love triangle, **all three pairs** must be matched:

- Pair (i, j) matched with probability p
- Pair (i, k) matched with probability p
- Pair (j, k) matched with probability p

Since the coin flips for different pairs are **independent**:

$$P(T_{ijk} = 1) = p \cdot p \cdot p = p^3$$

Step 4: Apply linearity of expectation.

$$E[L] = E \left[\sum_{\{i,j,k\}} T_{ijk} \right] = \sum_{\{i,j,k\}} E[T_{ijk}] = \sum_{\{i,j,k\}} p^3 = \binom{n}{3} \cdot p^3$$

Note: Linearity of expectation applies regardless of whether the T_{ijk} are independent—and they are **not** independent, since triangles can share edges. This is precisely why linearity of expectation is so powerful here.

Step 5: Simplify.

$$E[L] = \frac{n(n-1)(n-2)}{6} \cdot p^3$$

Answer: $E[\text{love triangles}] = \frac{n(n-1)(n-2)}{6} p^3$
